ELEC361 Analog Communication Systems Midterm Exam **Fall 2020**

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Q1) $x(t) = \sum_{n=-\infty}^{\infty} \prod(\frac{t-4n}{2})$ is applied to the LTI system with $h(t) = 8Sinc(2t)Cos(4\pi t)$.

a) Find complex Fourier series expansion of $x(t)$. Find its Fourier transform and power spectral density (PSD) function. Sketch its PSD function.
b) Find transfer function of the system and sketch it. Find its energy spectral density (ESD) and autocorrelation functions. Find its energy by using both ESD and autocorrelation functions.
c) Find the PSD function of the output signal $y(t)$ and sketch it.
d) Is this system distortionless for the input signal $x(t)$? Why?

Q2) The message signal $m(t) = 5 - 5Sin(1000\pi t)$ modulates the carrier signal $c(t) = 100Cos(5000\pi t)$ to obtain DSB-SC-AM signal.

a) Find and sketch the modulated signal $y(t)$. Find Fourier transforms of $m(t)$ and $y(t)$. Sketch their spectra. Find the power and bandwidth of $y(t)$.

b) Find the signals at the input and output of the LPF in both time and frequency domains if the carrier signal is generated at the demodulator as $Cos[2\pi(2500 + \Delta f)t + \theta]$. What is the bandwidth of the LPF? Find the power of the signal at the output of the LPF.

c) Is the message signal obtained at the output of LPF? If not, What must the values of Δf and θ be to obtain the message signal?

Q3) The message signal $m(t) = 10Cos(2000\pi t) + 20Cos(4000\pi t)$ is multiplied by the square wave $x(t) = \sum_{n=-\infty}^{\infty} \prod(\frac{t-n10^{-4}}{0.510^{-4}})$ and passed through a bandpass filter (BPF) to obtain DSB-SC-AM signal.

a) If $km(t)Cos(60000\pi t)$ is obtained at the output of the BPF, what are the center frequency and bandwidth of the BPF? Sketch its transfer function. What is the value of k?
b) Skecth the spectrum at output of the BPF? Find the power of the modulated signal.
c) Sketch the block diagram of the demodulator. What is the bandwidth of the LPF ? Find the signals at the input and output of the LPF in both time and frequency domains. Sketch their spectra. Find the power of the signal at the output of the LPF.